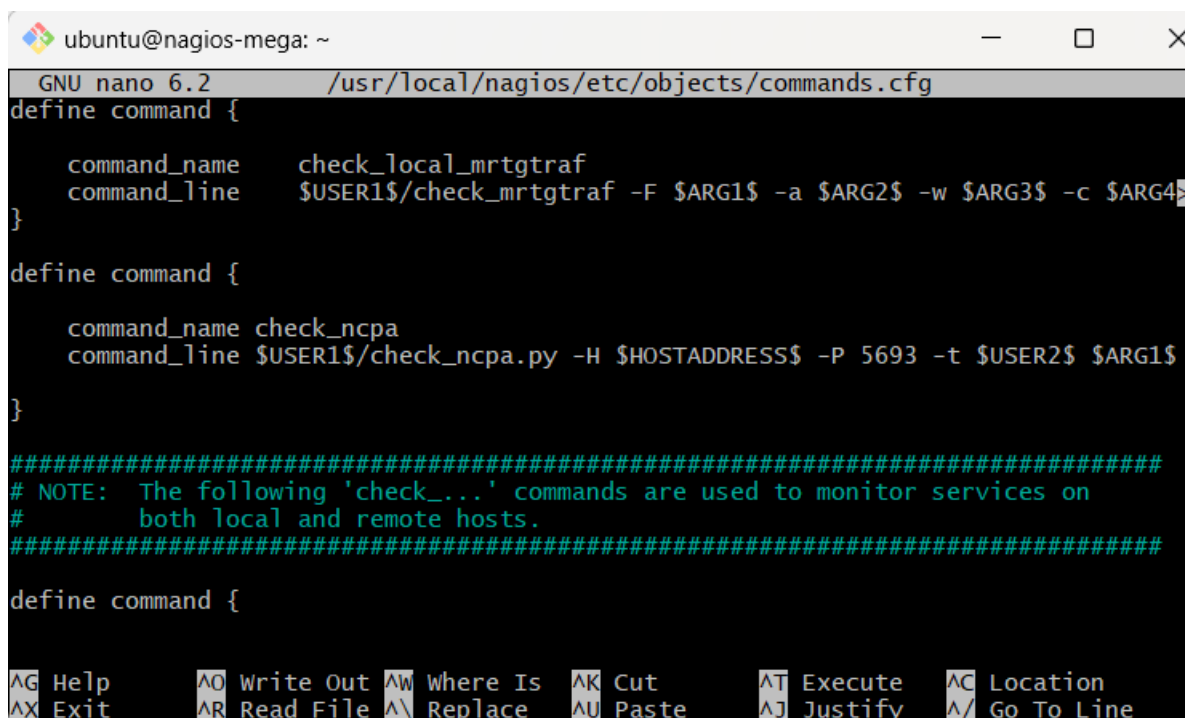


Nagios Mastery Part-4: Production-Grade End-to-End Observability of a Distributed Web Application Stack : Nginx Reverse Proxy, Flask Application, PostgreSQL Database and Two-Server Infrastructure(One of my most favourite and extensive practicals till this day)

Module 1: Initial NCPA discovery/checks for WEB-01 and Configuring WEB-01 Infrastructure Monitoring



```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/commands.cfg  
define command {  
    command_name    check_local_mrtgtraf  
    command_line    $USER1$/check_mrtgtraf -F $ARG1$ -a $ARG2$ -w $ARG3$ -c $ARG4$  
}  
  
define command {  
    command_name    check_ncpa  
    command_line    $USER1$/check_ncpa.py -H $HOSTADDRESS$ -P 5693 -t $USER2$ $ARG1$  
}  
  
#####  
# NOTE: The following 'check_...' commands are used to monitor services on  
#       both local and remote hosts.  
#####  
  
define command {  
  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location  
^X Exit      ^R Read File ^_ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg  
define host {  
  
    use linux-server  
    host_name WEB-01  
    alias Production Web Server  
    address 172.31.8.87  
    check_interval 1  
    retry_interval 1  
    max_check_attempts 1  
    notifications_enabled 1  
    notification_options d,u,r  
    check_period 24x7  
    notification_interval 30  
    notification_period 24x7  
    contact_groups admins  
  
}  
  
define service {  
  
[ Read 74 lines ]  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg  
  
}  
  
define service {  
  
    use generic-service  
    host_name WEB-01  
    service_description cpu-usage  
    check_command check_ncpa!-M cpu/percent -w 80 -c 90  
    check_interval 1  
    retry_interval 1  
    max_check_attempts 3  
    notifications_enabled 1  
    notification_options w,c,u,r  
    contact_groups admins  
  
}  
  
define service {  
  
    use generic-service
```

```
ubuntu@nagios-mega: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg
define service {

    use generic-service
    host_name WEB-01
    service_description memory-usage
    check_command check_ncpa!-M memory/virtual/percent -w 80 -c 90
    check_interval 1
    retry_interval 1
    max_check_attempts 3
    notifications_enabled 1
    notification_options w,c,u,r
    contact_groups admins
}

define service {

    use generic-service
    host_name WEB-01
    service_description root-disk-usage
    check_command check_ncpa!-M 'disk/logical/|/used_percent' -w 80 -c 90

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
ubuntu@nagios-mega: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg

define service {

    use generic-service
    host_name WEB-01
    service_description root-disk-usage
    check_command check_ncpa!-M 'disk/logical/|/used_percent' -w 80 -c 90
    check_interval 1
    retry_interval 1
    max_check_attempts 3
    notifications_enabled 1
    notification_options w,c,u,r
    contact_groups admins
}

define service {

    use generic-service
    host_name WEB-01
    service_description nginx-check

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
ubuntu@nagios-mega: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg
notifications_enabled 1
notification_options w,c,u,r
contact_groups admins
}

define service {

    use generic-service
    host_name WEB-01
    service_description nginx-check
    check_command check_ncpa!-M services -q "service=nginx,status=running,check=1" ->
    check_interval 1
    retry_interval 1
    max_check_attempts 3
    notifications_enabled 1
    notification_options w,c,u,r
    contact_groups admins
}

AG Help      AO Write Out  AW Where Is  AK Cut       AT Execute   AC Location
AX Exit      AR Read File  AL Replace   AU Paste     AJ Justify   AA Go To Line
```

Nagios

General

Home

Documentation

Current Status

Tactical Overview

Map

Hosts

Services

Host Groups

Summary

Grid

Service Groups

Summary

Grid

Problems

Services (Unhandled)

Hosts (Unhandled)

Network Outages

Quick Search:

Reports

Availability

Trends

Alerts

History

Summary

Histogram

Notifications

Event Log

System

Comments

Downtime

Process Info

Performance Info

Current Network Status

Last Updated: Thu Sep 17 08:12:17 UTC 2026

Updated every 90 seconds

Nagios® Core™ 4.5.10 - www.nagios.org

Logged in as nagiosadmin

View History For all Hosts

View Notifications For All Hosts

View Host Status Detail For All Hosts

Host Status Totals

Up 2 Down 0 Unreachable 0 Pending 0

All Problems All Types

0 2

Service Status Totals

OK 11 Warning 0 Unknown 0 Critical 1 Pending 0

All Problems All Types

1 12

Service Status Details For All Hosts

Limit Results: 100

Host	Service	Status	Last Check	Duration	Attempt	Status Information
WEB-01	cpu-usage	OK	09-17-2026 08:11:23	0d 0h 7m 54s	1/3	OK: Percent was 0.00 %, 0.00 %
	memory-usage	OK	09-17-2026 08:12:07	0d 0h 8m 10s	1/3	OK: Percent was 45.60 %
	nginx-check	OK	09-17-2026 08:11:44	0d 0h 7m 33s	1/3	OK: nginx is running
	root-disk-usage	OK	09-17-2026 08:11:53	0d 0h 6m 24s	1/3	OK: Used_percent was 27.60 %
localhost	Current Load	OK	09-17-2026 08:08:39	0d 20h 51m 37s	1/4	OK - load average: 0.00, 0.00, 0.00
	Current Users	OK	09-17-2026 08:09:17	0d 20h 50m 59s	1/4	USERS OK - 1 users currently logged in
	HTTP	OK	09-17-2026 08:09:54	0d 20h 50m 22s	1/4	HTTP OK: HTTP/11 200 OK - 10945 bytes in 0.000 second response time
	PING	OK	09-17-2026 08:10:32	0d 20h 49m 44s	1/4	PING OK - Packet loss = 0%, RTA = 0.04 ms
	Root Partition	OK	09-17-2026 08:11:09	0d 20h 49m 7s	1/4	DISK OK - free space: / 5313 MB (88.64% inode=92%):
	SSH	OK	09-17-2026 08:11:47	0d 20h 48m 29s	1/4	SSH OK - OpenSSH_8.9p1 Ubuntu-Subuntu0.17 (protocol 2.0)
	Swap Usage	CRITICAL	09-17-2026 08:07:24	0d 20h 44m 52s	4/4	SWAP CRITICAL - 0% free (0 MB out of 0 MB) - Swap is either disabled, not present, or of zero size.
	Total Processes	OK	09-17-2026 08:08:02	0d 20h 47m 14s	1/4	PROCS OK: 44 processes with STATE = RSZDT

Results 1 - 12 of 12 Matching Services

Module 2: Initial NCPA discovery/checks for DB-01 and Configuring DB-01 Infrastructure Monitoring

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg  
contact_groups admins  
}  
  
define host {  
  
    use linux-server  
    host_name DB-01  
    alias Production Database Server  
    address 172.31.4.174  
    check_interval 1  
    retry_interval 1  
    max_check_attempts 3  
    check_period 24x7  
    notifications_enabled 1  
    notification_options d,u,r  
    notification_period 24x7  
    contact_groups admins  
}  
  
define service {  
  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/commands.cfg  
  
define command {  
  
    command_name    check_local_mrtgtraf  
    command_line    $USER1$/check_mrtgtraf -F $ARG1$ -a $ARG2$ -w $ARG3$ -c $ARG4$  
}  
  
define command {  
  
    command_name check_ncpa  
    command_line $USER1$/check_ncpa.py -H $HOSTADDRESS$ -P 5693 -t $ARG1$ $ARG2$  
}  
  
#####  
# NOTE: The following 'check_...' commands are used to monitor services on  
#       both local and remote hosts.  
#####  
  
define command {  
  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg  
}  
  
define service {  
    use generic-service  
    host_name DB-01  
    service_description cpu-usage  
    check_command check_ncpa!$USER3$!-M cpu/percent -w 80 -c 90  
    check_interval 1  
    retry_interval 1  
    max_check_attempts 3  
    notifications_enabled 1  
    notification_options w,c,u,r  
    contact_groups admins  
}  
  
define service {  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location  
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg  
}  
  
define service {  
    use generic-service  
    host_name DB-01  
    service_description memory-usage  
    check_command check_ncpa!$USER3$!-M memory/virtual/percent -w 80 -c 90  
    check_interval 1  
    retry_interval 1  
    max_check_attempts 3  
    notifications_enabled 1  
    notification_options w,c,u,r  
    contact_groups admins  
}  
  
define service {  
    use generic-service  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location  
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg  
}  
  
define service {  
    use generic-service  
    host_name DB-01  
    service_description root disk usage  
    check_command check_ncpa!$USER3$!-M 'disk/logical/|/used_percent' -w 80 -c 90  
    check_interval 1  
    retry_interval 1  
    max_check_attempts 3  
    notifications_enabled 1  
    notification_options w,c,u,r  
    contact_groups admins  
}  
  
define service {  
    use generic-service
```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^_ Go To Line

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg  
}  
  
define service {  
    use generic-service  
    host_name DB-01  
    service_description NCPA agent version  
    check_command check_ncpa!$USER3$!-M system/agent_version  
    check_interval 1  
    retry_interval 1  
    max_check_attempts 3  
    notifications_enabled 1  
    notification_options w,c,u,r  
    contact_groups admins  
}  
  

```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^_ Go To Line

65.0.139.20/nagios/

Machine LearningNeural Networks ar...Density-based clust...Fisher LDA-StanfordCompetitive Progra...CVS in Unsupervise...Red Hat AcademySystem Design Primer

All Bookmarks

Nagios

General

Home

Documentation

Current Status

Tactical Overview

Map

Hosts

Services

Host Groups

Summary

Grid

Service Groups

Summary

Grid

Problems

Services (Unhandled)

Hosts (Unhandled)

Network Outages

Quick Search:

Reports

Availability

Trends

Alerts

History

Summary

Histogram

Notifications

Event Log

System

Comments

Downtime

Process Info

Performance Info

Current Network Status

Last Updated: Thu Sep 17 09:52:36 UTC 2026

Updated every 90 seconds

Nagios® Core™ 4.5.10 - www.nagios.org

Logged in as nagiosadmin

View History For all hosts

View Notifications For All Hosts

View Host Status Detail For All Hosts

Host Status Totals

Up Down Unreachable Pending

3 0 0 0

All Problems All Types

0 3

Service Status Totals

Ok Warning Unknown Critical Pending

15 0 0 1 0

All Problems All Types

1 16

Service Status Details For All Hosts

Limit Results: 100

Host	Service	Status	Last Check	Duration	Attempt	Status Information
DB-01	NCPA agent version	OK	09-17-2026 09:52:07	0d 0h 14m 29s	1/3	OK: Agent_version was 3.5.0
	cpu-usage	OK	09-17-2026 09:52:33	0d 0h 14m 3s	1/3	OK: Percent was 2.00 %, 0.00 %
	memory-usage	OK	09-17-2026 09:52:12	0d 0h 14m 24s	1/3	OK: Percent was 21.00 %
	root disk usage	OK	09-17-2026 09:52:27	0d 0h 14m 9s	1/3	OK: Used_percent was 25.00 %
WEB-01	cpu-usage	OK	09-17-2026 09:52:16	0d 0h 14m 20s	1/3	OK: Percent was 0.00 %, 0.00 %
	memory-usage	OK	09-17-2026 09:51:53	0d 0h 14m 43s	1/3	OK: Percent was 44.40 %
	nginx-check	OK	09-17-2026 09:52:15	0d 0h 14m 21s	1/3	OK: nginx is running
	root-disk-usage	OK	09-17-2026 09:51:39	0d 0h 14m 57s	1/3	OK: Used_percent was 27.60 %
localhost	Current Load	OK	09-17-2026 09:49:34	0d 22h 31m 56s	1/4	OK - load average: 0.00, 0.00, 0.00
	Current Users	OK	09-17-2026 09:50:29	0d 22h 31m 18s	1/4	USERS OK - 1 users currently logged in
	HTTP	OK	09-17-2026 09:51:24	0d 22h 30m 41s	1/4	HTTP OK: HTTP/1.1 200 OK - 10945 bytes in 0.000 second response time
	PING	OK	09-17-2026 09:52:19	0d 22h 30m 3s	1/4	PING OK - Packet loss = 0%, RTT = 0.04 ms
	Root Partition	OK	09-17-2026 09:49:52	0d 22h 29m 26s	1/4	DISK OK - free space / 5321 MB (60.76% inode=92%):
	SSH	OK	09-17-2026 09:50:47	0d 22h 28m 48s	1/4	SSH OK - OpenSSH_8.9p1 Ubuntu-3ubuntu0.17 (protocol 2.0)
	Swap Usage	CRITICAL	09-17-2026 09:51:42	0d 22h 25m 11s	4/4	SWAP CRITICAL - 0% free IO MB out of 0 MB! - Swap is either disabled, not present, or of zero size.
	Total Processes	OK	09-17-2026 09:47:37	0d 22h 27m 33s	1/4	PROCS OK: 42 processes with STATE = RSDZT

Results 1 - 16 of 16 Matching Services

Page Tour

Stress test for Mail check:

65.0.139.20/nagios/

Machine LearningNeural Networks ar...Density-based clust...Fisher LDA-StanfordCompetitive Progra...CVS in Unsupervise...Red Hat AcademySystem Design Primer

All Bookmarks

Nagios

General

Home

Documentation

Current Status

Tactical Overview

Map

Hosts

Services

Host Groups

Summary

Grid

Service Groups

Summary

Grid

Problems

Services (Unhandled)

Hosts (Unhandled)

Network Outages

Quick Search:

Reports

Availability

Trends

Alerts

History

Summary

Histogram

Notifications

Event Log

System

Comments

Downtime

Process Info

Performance Info

Service Information

Last Updated: Thu Sep 17 09:55:33 UTC 2026

Updated every 90 seconds

Nagios® Core™ 4.5.10 - www.nagios.org

Logged in as nagiosadmin

View Information For This Host

View Status Detail For This Host

View Alert History For This Service

View Trends For This Service

View Alert Histogram For This Service

View Availability Report For This Service

View Notifications For This Service

Service

nginx-check

On Host

Production Web Server

(WEB-01)

Member of

No servicegroups.

172.31.8.87

Service State Information

Current Status: CRITICAL (for 0d 0h 0m 18s)

Status Information: CRITICAL: nginx is stopped (should be running)

Performance Data:

Current Attempt: 1/3 (SOFT state)

Last Check Time: 09-17-2026 09:55:15

Check Type: ACTIVE

Check Latency / Duration: 0:000 / 0:118 seconds

Next Scheduled Check: 09-17-2026 09:56:15

Last State Change: 09-17-2026 09:55:15

Last Notification: N/A (notification 0)

Is This Service Flapping? NO (414% state change)

In Scheduled Downtime? NO

Last Update: 09-17-2026 09:55:24 (0d 0h 0m 9s ago)

Active Checks: ENABLED

Passive Checks: ENABLED

Obsessing: ENABLED

Notifications: ENABLED

Event Handler: ENABLED

Flap Detection: ENABLED

Service Commands

Disable active checks of this service

Re-schedule the next check of this service

Submit passive check result for this service

Stop accepting passive checks for this service

Stop obsessing over this service

Disable notifications for this service

Delay next service notification

Send custom service notification

Schedule downtime for this service

Disable event handler for this service

Disable flap detection for this service

Clear flapping state for this service

Service Comments

Add a new commentDelete all comments

Entry Time Author Comment Comment ID Persistent Type Expires Actions

This service has no comments associated with it

Page Tour

← → ↻ Not secure 65.0.139.20/nagios/

Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CVR in Unsupervise... Red Hat Academy System Design Primer All Bookmarks

Nagios

General

- Home
- Documentation
- Current Status**
- Tactical Overview
- Map
- Hosts
- Services
- Host Groups
- Summary
- Grid
- Service Groups
- Summary
- Grid
- Problems
- Services (Unhandled)
- Hosts (Unhandled)
- Network Outages
- Quick Search:

Reports

- Availability
- Trends
- Alerts
- History
- Summary
- Histogram
- Notifications
- Event Log
- System**
- Comments
- Downtime
- Process Info
- Performance Info

Service Information

Last Updated: Thu Sep 17 09:57:26 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin

View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service

Service
nginx-check
On Host
Production Web Server
(WEB-01)

Member of
No servicegroups.

172.31.8.87

Service State Information

Current Status: **CRITICAL** (for 0d 0h 0m 12s)
Status Information: CRITICAL: nginx is stopped (should be running)
Performance Data:
Current Attempt: 3/3 (HARD state)
Last Check Time: 09-17-2026 09:57:16
Check Type: ACTIVE
Check Latency / Duration: 0:000 / 0:134 seconds
Next Scheduled Check: 09-17-2026 09:58:16
Last State Change: 09-17-2026 09:57:16
Last Notification: 09-17-2026 09:57:16 (notification 1)
Is This Service Flapping? **NO** (10.26% state change)
In Scheduled Downtime? **NO**
Last Update: 09-17-2026 09:57:24 (0d 0h 0m 4s ago)

Active Checks: **ENABLED**
Passive Checks: **ENABLED**
Obsessing: **ENABLED**
Notifications: **ENABLED**
Event Handler: **ENABLED**
Flap Detection: **ENABLED**

Service Commands

- ✗ Disable active checks of this service
- ⌚ Re-schedule the next check of this service
- ❓ Submit passive check result for this service
- ✗ Stop accepting passive checks for this service
- ✗ Stop obsessing over this service
- ✓ Acknowledge this service problem
- ✗ Disable notifications for this service
- ⌚ Delay next service notification
- ✉ Send custom service notification
- ⌚ Schedule downtime for this service
- ✗ Disable event handler for this service
- ✗ Disable flap detection for this service
- ✗ Clear flapping state for this service

Service Comments

🔗 Add a new comment 🗑 Delete all comments

Entry Time	Author	Comment	Comment ID	Persistent	Type	Expires	Actions
This service has no comments associated with it							

Page Tour

≡ Gmail in:spam X

1 of 124 < > 🗑

**** PROBLEM Service Alert: Production Web Server/nginx-check is CRITICAL **** Spam x

📧 sreevishnu0407@gmail.com 📧 amazonses.com 15:27 (0 minutes ago) ☆ 🗑 ↶

to me

Why is this message in spam? This message is similar to messages that were identified as spam in the past.

Report as not spam ⓘ

***** Nagios *****

Notification Type: PROBLEM

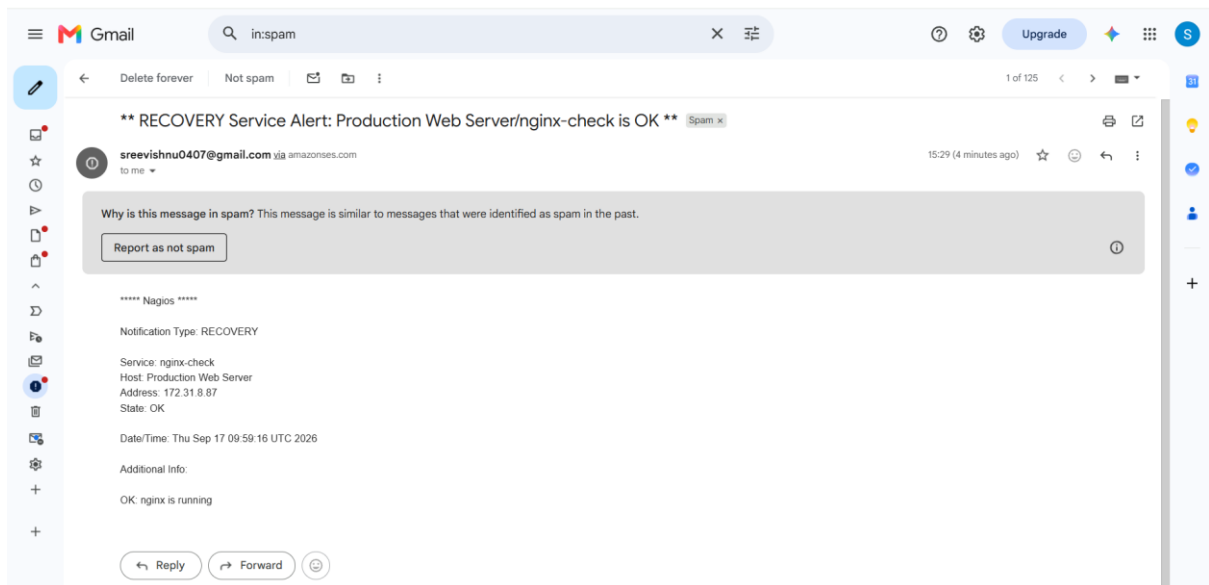
Service: nginx-check
Host: Production Web Server
Address: 172.31.8.87
State: CRITICAL

Date/Time: Thu Sep 17 09:57:16 UTC 2026

Additional Info:

CRITICAL: nginx is stopped (should be running)

Stress testing done



Recovery loop is perfect too

Module 3: PostgreSQL Setup in DB-01

```
ubuntu@DB-01: ~  
ubuntu@DB-01:~$ sudo apt install postgresql postgresql-contrib -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  libcommon-sense-perl libjson-perl libjson-xs-perl libllvm14  
  libtypes-serialiser-perl postgresql-14 postgresql-client-14  
  postgresql-client-common postgresql-common ssl-cert sysstat  
Suggested packages:  
  postgresql-doc postgresql-doc-14 isag  
The following NEW packages will be installed:  
  libcommon-sense-perl libjson-perl libjson-xs-perl libllvm14  
  libtypes-serialiser-perl postgresql postgresql-14 postgresql-client-14  
  postgresql-client-common postgresql-common postgresql-contrib ssl-cert  
  sysstat  
0 upgraded, 13 newly installed, 0 to remove and 8 not upgraded.  
Need to get 42.4 MB of archives.  
After this operation, 161 MB of additional disk space will be used.  
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libcommon-  
sense-perl amd64 3.75-2build1 [21.1 kB]  
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libjson-p-  
erl all 4.04000-1 [81.8 kB]  
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libtypes-  
serialiser-perl all 1.01-1 [11.6 kB]
```

```
ubuntu@DB-01: ~  
ubuntu@DB-01:~$ sudo -u postgres psql -c "Select 1;"  
could not change directory to "/home/ubuntu": Permission denied  
?column?  
-----  
      1  
(1 row)  
  
ubuntu@DB-01:~$ psql  
psql: error: connection to server on socket "/var/run/postgresql/.s.PGSQL.5432"  
failed: FATAL:  role "ubuntu" does not exist  
ubuntu@DB-01:~$ sudo -u postgres psql  
could not change directory to "/home/ubuntu": Permission denied  
psql (14.24 (Ubuntu 14.24-0ubuntu0.22.04.1))  
Type "help" for help.  
  
postgres=# \q  
ubuntu@DB-01:~$ sudo -u postgres psql  
could not change directory to "/home/ubuntu": Permission denied  
psql (14.24 (Ubuntu 14.24-0ubuntu0.22.04.1))  
Type "help" for help.  
  
postgres=# create database vish_db  
postgres=# ;  
CREATE DATABASE
```

```
postgres=# create database vish_db  
postgres=# ;  
CREATE DATABASE  
postgres=# create user nagios_monitor with login password 'vishnu123'  
postgres=# ;  
CREATE ROLE  
postgres=# grant connect on database vish_db on nagios_monitor;  
ERROR:  syntax error at or near "on"  
LINE 1: grant connect on database vish_db on nagios_monitor;  
                                         ^  
postgres=# grant connect on database vish_db to nagios_monitor;  
GRANT  
postgres=# show hba_file;  
          hba_file  
-----  
/etc/postgresql/14/main/pg_hba.conf  
(1 row)  
  
postgres=# \a  
Output format is unaligned.  
postgres=# \q  
ubuntu@DB-01:~$ sudo nano /etc/postgresql/14/main/pg_hba.conf  
ubuntu@DB-01:~$ sudo systemctl reload postgresql  
ubuntu@DB-01:~$ psql -h 127.0.0.1 -U nagios_monitor -d vish_db -w
```

```
ubuntu@DB-01: ~  
GNU nano 6.2 /etc/postgresql/14/main/pg_hba.conf  
# maintenance (custom daily cronjobs, replication, and similar tasks).  
#  
# Database administrative login by Unix domain socket  
local all postgres peer  
# TYPE DATABASE USER ADDRESS METHOD  
# "local" is for Unix domain socket connections only  
local all all peer  
# IPv4 local connections:  
host all all 127.0.0.1/32 scram-sha-256  
# IPv6 local connections:  
host all all ::1/128 scram-sha-256  
# Allow replication connections from localhost, by a user with the  
# replication privilege.  
local replication all peer  
host replication all 127.0.0.1/32 scram-sha-256  
host replication all ::1/128 scram-sha-256  
host vish_db nagios_monitor 172.31.14.204/32 scram-sha-256  
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location  
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line
```

```
ubuntu@DB-01: ~  
GNU nano 6.2 /etc/postgresql/14/main/postgresql.conf  
#-----  
# CONNECTIONS AND AUTHENTICATION  
#-----  
# - Connection Settings -  
listen_addresses = '*' # what IP address(es) to listen on;  
# comma-separated list of addresses;  
# defaults to 'localhost'; use '*' for  
# (change requires restart)  
port = 5432 # (change requires restart)  
max_connections = 100 # (change requires restart)  
#superuser_reserved_connections = 3 # (change requires restart)  
unix_socket_directories = '/var/run/postgresql' # comma-separated list of direc  
# (change requires restart)  
#unix_socket_group = '' # (change requires restart)  
#unix_socket_permissions = 0777 # begin with 0 to use octal notation  
# (change requires restart)  
#bonjour = off # advertise server via Bonjour  
# (change requires restart)
```

```
ubuntu@nagios-mega:~$ /usr/lib/nagios/plugins/check_pgsql -H 172.31.4.174 -P 5432  
-d vish_db -l nagios_monitor -p "vishnu123"  
OK - database vish_db (0.021061 sec.)|time=0.021061s;2.000000;8.000000;0.000000  
ubuntu@nagios-mega:~$
```

PostgreSQL test from Nagios-Master succeeded

Module 4: Custom Python PostgreSQL Application-Health Monitoring

```
ubuntu@DB-01: ~  
ubuntu@DB-01:~$ sudo -u postgres psql vish_db  
could not change directory to "/home/ubuntu": Permission denied  
psql (14.24 (Ubuntu 14.24-0ubuntu0.22.04.1))  
Type "help" for help.  
  
vish_db=# \dt  
          List of relations  
Schema |   Name   | Type | Owner  
-----+-----+-----+-----  
public | customers | table | postgres  
(1 row)  
  
vish_db=# select * from customers  
vish_db=# ;  
 id | name  | email  
----+-----+-----  
  1 | vishnu | vish123@gmail.com  
  2 | hari   | hari123@gmail.com  
  3 | anisu  | SA@gmail.com  
(3 rows)  
  
vish_db=# GRANT SELECT ON TABLE customers to nagios_monitor;  
GRANT  
vish_db=# |
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/libexec/check_postgres_health.py  
#!/usr/bin/env python3  
  
import sys  
import time  
import os  
import psycopg2  
  
HOST=sys.argv[1]  
DATABASE=sys.argv[2]  
USER=sys.argv[3]  
WARNING=float(sys.argv[4])  
CRITICAL=float(sys.argv[5])  
  
os.environ["PGPASSFILE"]=" /usr/local/nagios/etc/.pgpass"  
  
try:  
    start = time.time()  
  
    conn = psycopg2.connect(  
        host=HOST,  
[ Read 53 lines ]  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location  
^X Exit      ^R Read File ^N Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/libexec/check_postgres_health.py  
try:  
    start = time.time()  
  
    conn = psycopg2.connect(  
        host=HOST,  
        database=DATABASE,  
        port=5432,  
        user=USER,  
        connect_timeout=5  
    )  
    curr = conn.cursor()  
    curr.execute("select count(*) from customers;")  
    count=curr.fetchone()[0]  
  
    curr.close()  
    conn.close()  
  
    response_time = time.time()-start  
  
    if response_time >= CRITICAL:
```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^_ Go To Line

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/libexec/check_postgres_health.py  
  
    if response_time >= CRITICAL:  
        print(f"CRITICAL Database query time={response_time:.2f},customers={count}")  
        sys.exit(2)  
  
    if response_time >= WARNING:  
        print(f"WARNING Database query time={response_time:.2f},customers={count}")  
        sys.exit(1)  
  
    print(f"OK Database query time={response_time:.2f},customers={count}")  
    sys.exit(0)  
  
except psycopg2.Error as e:  
    print(f"CRITICAL postgresql query failed: {e}")  
    sys.exit(2)  
  
except Exception as e:  
    print(f"UNKNOWN some unknown plugin error: {e}")  
    sys.exit(3)
```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^_ Go To Line

```
ubuntu@nagios-mega: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/commands.cfg
define command {
    command_name    check_local_mrtgtraf
    command_line    $USER1$/check_mrtgtraf -F $ARG1$ -a $ARG2$ -w $ARG3$ -c $ARG4$ -e $ARG5$
}

define command {
    command_name    check_ncpa
    command_line    $USER1$/check_ncpa.py -H $HOSTADDRESS$ -P 5693 -t $ARG1$ $ARG2$
}

define command {
    command_name    check_postgres_health
    command_line    $USER1$/check_postgres_health.py $ARG1$ $ARG2$ $ARG3$ $ARG4$ $ARG5$
}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^_ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
ubuntu@nagios-mega: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg
contact_groups admins
}

define service {
    use generic-service
    host_name DB-01
    service_description PostgreSQL Application Health
    check_command check_postgres_health!$HOSTADDRESS$!vish_db!nagios_monitor!2!5
    check_interval 1
    retry_interval 1
    max_check_attempts 3
    check_period 24x7
    notifications_enabled 1
    notification_period 24x7
    notification_options w,c,u,r
    contact_groups admins
}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^_ Replace   ^U Paste     ^J Justify
```


Nagios
 Last Updated: Sat Sep 19 06:53:47 UTC 2026
 Updated every 90 seconds
 Nagios® Core™ 4.5.10 - www.nagios.org
 Logged in as nagiosadmin

Current Network Status

Host Status Totals

Up	Down	Unreachable	Pending
3	0	0	0

All Problems: 0, All Types: 3

Service Status Totals

Ok	Warning	Unknown	Critical	Pending
15	1	0	1	0

All Problems: 2, All Types: 17

Service Status Details For All Hosts

Host	Service	Status	Last Check	Duration	Attempt	Status Information
DB-01	NCPA agent version	OK	09-19-2026 06:53:21	1d 21h 15m 40s	1/3	OK: Agent_version was 3.5.0
	PostgreSQL Application Health	WARNING	09-19-2026 06:53:09	0d 0h 0m 31s	1/3	WARNING Database query time=0.02,customers=3
	cpu-usage	OK	09-19-2026 06:53:09	0d 20h 12m 40s	1/3	OK: Percent was 0.00 %, 0.00 %
	memory-usage	OK	09-19-2026 06:53:19	1d 21h 15m 38s	1/3	OK: Percent was 21.90 %
	root-disk-usage	OK	09-19-2026 06:53:34	1d 21h 15m 20s	1/3	OK: Used_percent was 31.90 %
WEB-01	cpu-usage	OK	09-19-2026 06:53:32	0d 0h 37m 15s	1/3	OK: Percent was 0.00 %, 0.00 %
	memory-usage	OK	09-19-2026 06:53:02	0d 0h 37m 45s	1/3	OK: Percent was 45.00 %
	nginx-check	OK	09-19-2026 06:53:33	0d 0h 37m 14s	1/3	OK: nginx is running
	root-disk-usage	OK	09-19-2026 06:53:01	0d 0h 37m 46s	1/3	OK: Used_percent was 29.40 %
localhost	Current Load	OK	09-19-2026 06:52:40	2d 19h 33m 7s	1/4	OK - load average: 0.00, 0.00, 0.00
	Current Users	OK	09-19-2026 06:53:14	2d 19h 32m 29s	1/4	USERS OK - 1 users currently logged in
	HTTP	OK	09-19-2026 06:48:48	2d 19h 31m 52s	1/4	HTTP OK: HTTP/11 200 OK - 10945 bytes in 0.000 second response time
	PING	OK	09-19-2026 06:49:21	2d 19h 31m 14s	1/4	PING OK - Packet loss = 0%, RTT = 0.04 ms
	Root Partition	OK	09-19-2026 06:51:44	2d 19h 30m 37s	1/4	DISK OK - free space: / 5247 MiB (67.79% inode=97%)
	SSH	OK	09-19-2026 06:52:18	2d 19h 29m 59s	1/4	SSH OK - OpenSSH_8.9p1 Ubuntu:suburtil@ (protocol 2.0)
	Swap Usage	CRITICAL	09-19-2026 06:52:51	2d 19h 26m 22s	4/4	SWAP CRITICAL - 0% free (0 MB out of 0 MB) - Swap is either disabled, not present, or of zero size.
	Total Processes	OK	09-19-2026 06:53:25	2d 19h 28m 44s	1/4	PROCS OK: 42 processes with STATE = RSDT

Results 1 - 17 of 17 Matching Services

Gmail
 in:spam

**** PROBLEM Service Alert: Production Database Server/PostgreSQL Application Health is WARNING ****

from: sreevishnu0407@gmail.com via amazonses.com
 to: me

12:25 (0 minutes ago)

Why is this message in spam? This message is similar to messages that were identified as spam in the past.

[Report as not spam](#)

***** Nagios *****

Notification Type: PROBLEM

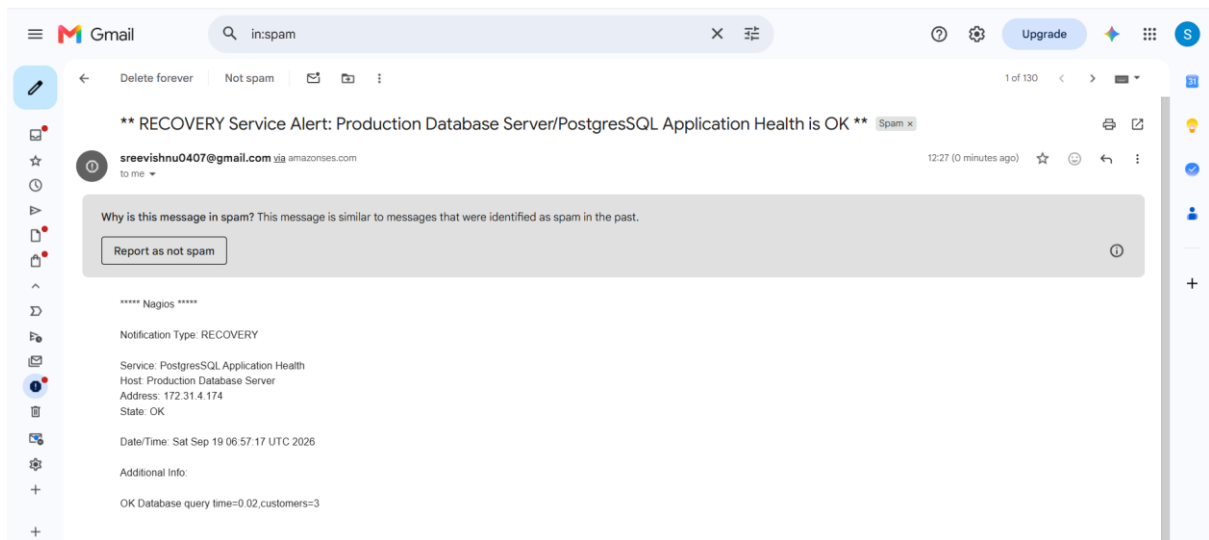
Service: PostgreSQL Application Health
 Host: Production Database Server
 Address: 172.31.4.174
 State: WARNING

Date/Time: Sat Sep 19 06:55:16 UTC 2026

Additional Info:

WARNING Database query time=0.02,customers=3

There we go, WARNING simulated



After changing to normal thresholds, we are back to recovery (whole pipeline verified)

Module 5: WEB-01 ↔ DB-01 Application Integration (Main)

(One of the best modules of my DevOps Mastery)

```

ubuntu@DB-01: ~
GNU nano 6.2 /etc/postgresql/14/main/pg_hba.conf
#
# Database administrative login by Unix domain socket
local all postgres peer
# TYPE DATABASE USER ADDRESS METHOD
# "local" is for Unix domain socket connections only
local all all peer
# IPv4 local connections:
host all all 127.0.0.1/32 scram-sha-256
# IPv6 local connections:
host all all ::1/128 scram-sha-256
# Allow replication connections from localhost, by a user with the
# replication privilege.
local replication all peer
host replication all 127.0.0.1/32 scram-sha-256
host replication all ::1/128 scram-sha-256
host vish_db nagios_monitor 172.31.14.204/32 scram-sha-256
host vish_db nagios_monitor 172.31.8.87/32 scram-sha-256
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line
  
```

```
ubuntu@nagios-check2: ~  
System load: 0.05      Processes: 119  
Usage of /: 28.7% of 7.57GB Users logged in: 0  
Memory usage: 28%      IPv4 address for ens5: 172.31.8.87  
Swap usage: 0%  
  
* Ubuntu Pro delivers the most comprehensive open source security and  
compliance features.  
  
https://ubuntu.com/aws/pro  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
Last login: Fri Sep 18 09:02:05 2026 from 210.89.61.42  
ubuntu@nagios-check2:~$ sudo hostname -i  
172.31.8.87 fe80::8ff:f9ff:fe06:b32f  
ubuntu@nagios-check2:~$ nc -zv 172.31.4.174 5432  
Connection to 172.31.4.174 5432 port [tcp/postgresql] succeeded!  
ubuntu@nagios-check2:~$
```

```
ubuntu@nagios-check2: ~  
ubuntu@nagios-check2:~$ sudo apt install python3-psycopg2 -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  libpq5  
Suggested packages:  
  python-psycopg2-doc  
The following NEW packages will be installed:  
  libpq5 python3-psycopg2  
0 upgraded, 2 newly installed, 0 to remove and 8 not upgraded.  
Need to get 293 kB of archives.  
After this operation, 926 kB of additional disk space will be used.  
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpq5 amd64 14.24-0ubuntu0.22.04.1 [157 kB]  
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 python3-psycopg2 amd64 2.9.2-1build2 [136 kB]  
Fetched 293 kB in 0s (9556 kB/s)  
Selecting previously unselected package libpq5:amd64.  
(Reading database ... 66451 files and directories currently installed.)  
Preparing to unpack .../libpq5_14.24-0ubuntu0.22.04.1_amd64.deb ...  
Unpacking libpq5:amd64 (14.24-0ubuntu0.22.04.1) ...  
Selecting previously unselected package python3-psycopg2.  
Preparing to unpack .../python3-psycopg2_2.9.2-1build2_amd64.deb ...
```

```
ubuntu@nagios-check2: ~  
Unpacking libpq5:amd64 (14.24-0ubuntu0.22.04.1) ...  
Selecting previously unselected package python3-psycopg2.  
Preparing to unpack .../python3-psycopg2_2.9.2-1build2_amd64.deb ...  
Unpacking python3-psycopg2 (2.9.2-1build2) ...  
Setting up libpq5:amd64 (14.24-0ubuntu0.22.04.1) ...  
Setting up python3-psycopg2 (2.9.2-1build2) ...  
Processing triggers for libc-bin (2.35-0ubuntu3.15) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@nagios-check2:~$ python3 -c "import psycopg2; c=psycopg2.connect(host='172.31.4.174',port=5432,database='vish_db',user='nagios_monitor',password='vishnu123');print('DB connection OK');c.close()"  
DB connection OK  
ubuntu@nagios-check2:~$ |
```

```
ubuntu@nagios-check2: ~  
ubuntu@nagios-check2:~$ sudo apt install python3-flask -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  javascript-common libjs-jquery python3-itsdangerous python3-pyinotify  
  python3-simplejson python3-werkzeug  
Suggested packages:  
  python3-asgiref python3-dotenv python-flask-doc python-pyinotify-doc  
  ipython3 python-werkzeug-doc python3-lxml python3-watchdog  
The following NEW packages will be installed:  
  javascript-common libjs-jquery python3-flask python3-itsdangerous  
  python3-pyinotify python3-simplejson python3-werkzeug  
0 upgraded, 7 newly installed, 0 to remove and 8 not upgraded.  
Need to get 684 kB of archives.  
After this operation, 2492 kB of additional disk space will be used.  
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 javascript-common all 11+nmu1 [5936 B]  
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libjs-jquery all 3.6.0+dfsg+~3.5.13-1 [321 kB]  
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 python3-itsdangerous all 2.1.0-1 [14.7 kB]  
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-werkzeug all 2.0.2+dfsg1-1ubuntu0.22.04.3 [181 kB]
```

```
ubuntu@nagios-check2: /opt/webapp
GNU nano 6.2 app.py
from flask import Flask, jsonify
import psycopg2
import os
from dotenv import load_dotenv

load_dotenv()

app = Flask(__name__)

DB_CONFIG = {
    "host": os.environ["DB_HOST"],
    "port": os.environ["DB_PORT"],
    "database": os.environ["DB_NAME"],
    "user": os.environ["DB_USER"],
    "password": os.environ["DB_PASSWORD"]
}

def get_db():
    return psycopg2.connect(**DB_CONFIG)

[ Read 76 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace  ^U Paste     ^J Justify  ^_ Go To Line
```

```
ubuntu@nagios-check2: /opt/webapp
GNU nano 6.2 app.py
@app.route("/")
def home():
    return """
    <html>
    <head><title>DevOps Demo stack</title></head>
    <body>
        <h1>DevOps Monitoring Demo</h1>
        <p>Application: ONLINE</p>
        <p>Database: CONNECTED</p>
        <p>Server: WEB-01</p>
    </body>
    </html>
    """

@app.route("/health")
def health():
    try:
        conn = get_db()
        curr = conn.cursor()
        curr.execute("SELECT 1;")
```

```
ubuntu@nagios-check2: /opt/webapp
GNU nano 6.2 app.py
@app.route("/health")
def health():
    try:
        conn = get_db()
        curr = conn.cursor()
        curr.execute("SELECT 1;")
        curr.fetchone()
        curr.close()
        conn.close()
        return jsonify({
            "status": "healthy",
            "database": "up"
        })
    except Exception as e:
        return jsonify({
            "status": "unhealthy",
            "database": "down",
            "error": str(e)
        }), 503

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

```
ubuntu@nagios-check2: /opt/webapp
GNU nano 6.2 app.py
@app.route("/api/status")
def status():
    try:
        conn = get_db()
        curr = conn.cursor()
        curr.execute("SELECT COUNT(*) FROM customers;")
        customers = curr.fetchone()[0]
        curr.close()
        conn.close()
        return jsonify({
            "application": "online",
            "database": "as up as Vishnu himself",
            "customers": customers
        })
    except Exception as e:
        return jsonify({
            "application": "online",
            "database": "maybe vishnu is down"
        }), 503

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

```
ubuntu@nagios-check2: /opt/webapp
GNU nano 6.2 .env
DB_HOST=172.31.4.174
DB_PORT=5432
DB_NAME=vish_db
DB_USER=nagios_monitor
DB_PASSWORD=vishnu123

[ Read 6 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^_ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

.env hidden file just to highlight its presence (but in production you never expose this)

```
ubuntu@nagios-check2: /opt/webapp
ubuntu@nagios-check2:/opt/webapp$ sudo nano .env
ubuntu@nagios-check2:/opt/webapp$ sudo cat /etc/nginx/sites-available/webapp
server {
    listen 80;
    server_name _;

    location / {
        proxy_pass http://127.0.0.1:5000;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
ubuntu@nagios-check2:/opt/webapp$ sudo ls -l /etc/nginx/sites-enabled/webapp
lrwxrwxrwx 1 root root 33 Sep 19 08:46 /etc/nginx/sites-enabled/webapp -> /etc/nginx/sites-available/webapp
ubuntu@nagios-check2:/opt/webapp$ sudo cat /etc/systemd/system/webapp.service
[Unit]
Description=DevOps Demo Flask Application
After=network.target

[Service]
User=www-data
WorkingDirectory=/opt/webapp
```

```
ubuntu@nagios-check2: /opt/webapp
proxy_pass http://127.0.0.1:5000;
proxy_set_header X-Real-IP $remote_addr;
}
}
ubuntu@nagios-check2:/opt/webapp$ sudo ls -l /etc/nginx/sites-enabled/webapp
lrwxrwxrwx 1 root root 33 Sep 19 08:46 /etc/nginx/sites-enabled/webapp -> /etc/n
ginx/sites-available/webapp
ubuntu@nagios-check2:/opt/webapp$ sudo cat /etc/systemd/system/webapp.service
[Unit]
Description=DevOps Demo Flask Application
After=network.target

[Service]
User=www-data
WorkingDirectory=/opt/webapp
ExecStart=/usr/bin/python3 /opt/webapp/app.py
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
ubuntu@nagios-check2:/opt/webapp$ |
```

Note: webapp. service is a systemd service unit we created to manage the Flask application as a background service- starting it automatically, restarting it if it fails, and running it under the www-data user.

Nagios Monitoring Files:

```
ubuntu@nagios-mega: ~
GNU nano 6.2 /usr/local/nagios/libexec/check_http_avail.py *
#!/usr/bin/env python3

import sys
import time
import requests

if len(sys.argv) != 4:
    print("UNKNOWN: Usage: check_http_availability.py <URL> <WARNING> <CRITI>")
    sys.exit(3)

url = sys.argv[1]
warning = float(sys.argv[2])
critical = float(sys.argv[3])

try:
    start = time.time()

    response = requests.get(url, timeout=10)

    response_time = time.time() - start

[ Read 47 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify
```

```
ubuntu@nagios-mega: ~
GNU nano 6.2 /usr/local/nagios/libexec/check_http_avail.py *
if response.status_code != 200:
    print(f"CRITICAL: HTTP {response.status_code}")
    sys.exit(2)

if response_time >= critical:
    print(f"CRITICAL: HTTP 200, response time={response_time:.2f}s")
    sys.exit(2)

if response_time >= warning:
    print(f"WARNING: HTTP 200, response time={response_time:.2f}s")
    sys.exit(1)

print(f"OK: HTTP 200, response time={response_time:.2f}s")
sys.exit(0)

except requests.exceptions.Timeout:
    print("CRITICAL: HTTP request timed out")
    sys.exit(2)

except requests.exceptions.RequestException as e:
    print(f"CRITICAL: HTTP request failed - {e}")
    sys.exit(2)

except Exception as e:
    print(f"UNKNOWN: Plugin error - {e}")
    sys.exit(3)

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify
```

```
ubuntu@nagios-mega: ~
GNU nano 6.2 /usr/local/nagios/libexec/check_web_health.py
#!/usr/bin/env python3

import sys
import time
import requests

if len(sys.argv) != 4:
    print("UNKNOWN: Usage: check_web_health.py <URL> <WARNING> <CRITICAL>")
    sys.exit(3)

url = sys.argv[1]
warning = float(sys.argv[2])
critical = float(sys.argv[3])

try:
    start = time.time()

    response = requests.get(url, timeout=10)

    response_time = time.time() - start

[ Read 61 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/libexec/check_web_health.py  
if response.status_code != 200:  
    print(f"CRITICAL: HTTP {response.status_code}")  
    sys.exit(2)  
  
try:  
    data = response.json()  
except ValueError:  
    print("CRITICAL: Invalid JSON response")  
    sys.exit(2)  
  
if data.get("status") != "healthy":  
    print("CRITICAL: Application status is not healthy")  
    sys.exit(2)  
  
if data.get("database") != "up":  
    print("CRITICAL: Database is down")  
    sys.exit(2)  
  
if response_time >= critical:  
    print(f"CRITICAL: Health check passed, response={response_time:.2f}s")  
    sys.exit(2)  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/libexec/check_web_health.py  
sys.exit(2)  
  
if response_time >= warning:  
    print(f"WARNING: Health check passed, response={response_time:.2f}s")  
    sys.exit(1)  
  
print(f"OK: Application healthy, database up, response={response_time:.2f}s")  
sys.exit(0)  
  
except requests.exceptions.Timeout:  
    print("CRITICAL: Health request timed out")  
    sys.exit(2)  
  
except requests.exceptions.RequestException as e:  
    print(f"CRITICAL: Health request failed - {e}")  
    sys.exit(2)  
  
except Exception as e:  
    print(f"UNKNOWN: Plugin error - {e}")  
    sys.exit(3)  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/libexec/check_web_app_health.py  
#!/usr/bin/env python3  
  
import sys  
import time  
import requests  
  
if len(sys.argv) != 4:  
    print("UNKNOWN: Usage: check_web_app_health.py <URL> <WARNING> <CRITICAL>")  
    sys.exit(3)  
  
url = sys.argv[1]  
warning = float(sys.argv[2])  
critical = float(sys.argv[3])  
  
try:  
    start = time.time()  
  
    response = requests.get(url, timeout=10)  
  
    response_time = time.time() - start  
  
[ Read 69 lines ]  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/libexec/check_web_app_health.py  
  
response_time = time.time() - start  
  
if response.status_code != 200:  
    print(f"CRITICAL: HTTP {response.status_code}")  
    sys.exit(2)  
  
try:  
    data = response.json()  
except ValueError:  
    print("CRITICAL: Invalid JSON response")  
    sys.exit(2)  
  
if data.get("application") != "online":  
    print("CRITICAL: Application is not online")  
    sys.exit(2)  
  
if data.get("database") != "as up as Vishnu himself":  
    print("CRITICAL: Application database is down")  
    sys.exit(2)  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/libexec/check_web_app_health.py  
if response_time >= warning:  
    print(f"WARNING: API healthy, response={response_time:.2f}s")  
    sys.exit(1)  
  
    print(  
        f"OK: API healthy, database up, customers={customers}, response={res  
    )  
    sys.exit(0)  
  
except requests.exceptions.Timeout:  
    print("CRITICAL: API request timed out")  
    sys.exit(2)  
  
except requests.exceptions.RequestException as e:  
    print(f"CRITICAL: API request failed - {e}")  
    sys.exit(2)  
  
except Exception as e:  
    print(f"UNKNOWN: Plugin error - {e}")  
    sys.exit(3)  
|  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/commands.cfg  
define command {  
  
    command_name check_http_avail  
    command_line $USER1$/check_http_avail.py $ARG1$ $ARG2$ $ARG3$  
}  
  
define command {  
  
    command_name check_web_health  
    command_line $USER1$/check_web_health.py $ARG1$ $ARG2$ $ARG3$  
}  
  
define command {  
  
    command_name check_web_app_health  
    command_line $USER1$/check_web_app_health.py $ARG1$ $ARG2$ $ARG3$  
}  
  
#####  
|  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
ubuntu@nagios-mega: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg
define service {
    use                generic-service
    host_name          WEB-01
    service_description Web Page Availability
    check_command       check_http_avail!http://172.31.8.87/!2!5
    check_interval     1
    retry_interval      1
    max_check_attempts 3
    check_period        24x7
    notifications_enabled 1
    notification_period 24x7
    notification_options w,c,u,r
    contact_groups      admins
}

define service {
    use                generic-service
    host_name          WEB-01
    service_description Web Application Health
    check_command       check_web_health!http://172.31.8.87/health!2!5
    check_interval     1
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify
```

```
ubuntu@nagios-mega: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg
    contact_groups      admins
}

define service {
    use                generic-service
    host_name          WEB-01
    service_description Web Application Health
    check_command       check_web_health!http://172.31.8.87/health!2!5
    check_interval     1
    retry_interval      1
    max_check_attempts 3
    check_period        24x7
    notifications_enabled 1
    notification_period 24x7
    notification_options w,c,u,r
    contact_groups      admins
}

define service {
    use                generic-service
    host_name          WEB-01
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify
```


Page Tour

Page Tour

←→↻⚠ Not secure43.204.230.200/nagios/

Machine LearningNeural Networks ar...Density-based clust...Fisher LDA-StanfordCompetitive Progra...CVB in Unsupervise...Red Hat AcademySystem Design Primer»All Bookmarks

Nagios

General

HomeDocumentationCurrent StatusTactical OverviewMapHostsServicesHost GroupsSummaryGridService GroupsSummaryGridProblemsServices (Unhandled)Hosts (Unhandled)Network OutagesQuick Search:ReportsAvailabilityTrendsAlertsHistorySummaryHistogramNotificationsEvent LogSystemCommentsDowntimeProcess InfoPerformance Info

Service Information

Last Updated: Sat Sep 19 10:48:06 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin

View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service

Service

Web Page Availability

On Host
Production Web Server
(WEB-01)

Member of
No servicegroups.

172.31.8.87

Service State Information

Current Status: OK (for 0d 0h 32m 4s)
Status information: OK: HTTP 200, response time=0.00s
Performance Data:
Current Attempt: 1/3 (HARD state)
Last Check Time: 09-19-2026 10:48:04
Check Type: ACTIVE
Check Latency / Duration: 0.000 / 0.132 seconds
Next Scheduled Check: 09-19-2026 10:49:04
Last State Change: 09-19-2026 10:16:04
Last Notification: 09-19-2026 10:16:04 (notification 0)
Is This Service Flapping? NO (0.00% state change)
In Scheduled Downtime? NO
Last Update: 09-19-2026 10:48:06 (0d 0h 0m 2s ago)
Active Checks: ENABLED
Passive Checks: ENABLED
Obsessing: ENABLED
Notifications: ENABLED
Event Handler: ENABLED
Flap Detection: ENABLED

Service Commands

✗ Disable active checks of this service

🕒 Re-schedule the next check of this service

❓ Submit passive check result for this service

✗ Stop accepting passive checks for this service

✗ Stop obsessing over this service

✗ Disable notifications for this service

📧 Send custom service notification

🕒 Schedule downtime for this service

✗ Disable event handler for this service

✗ Disable flap detection for this service

✗ Clear flapping state for this service

Service Comments

🗨 Add a new comment🗑 Delete all comments

Entry Time Author Comment Comment ID Persistent Type Expires Actions

This service has no comments associated with it

Page Tour

←→↻⚠ Not secure43.204.230.200/nagios/

Machine LearningNeural Networks ar...Density-based clust...Fisher LDA-StanfordCompetitive Progra...CVB in Unsupervise...Red Hat AcademySystem Design Primer»All Bookmarks

Nagios

General

HomeDocumentationCurrent StatusTactical OverviewMapHostsServicesHost GroupsSummaryGridService GroupsSummaryGridProblemsServices (Unhandled)Hosts (Unhandled)Network OutagesQuick Search:ReportsAvailabilityTrendsAlertsHistorySummaryHistogramNotificationsEvent LogSystemCommentsDowntimeProcess InfoPerformance Info

Service Information

Last Updated: Sat Sep 19 10:48:28 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin

View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service

Service

PostgreSQL Application Health

On Host
Production Database Server
(DB-01)

Member of
No servicegroups.

172.31.4174

Service State Information

Current Status: OK (for 0d 3h 51m 12s)
Status information: OK Database query time=0.02,customers=3
Performance Data:
Current Attempt: 1/3 (HARD state)
Last Check Time: 09-19-2026 10:48:16
Check Type: ACTIVE
Check Latency / Duration: 0.000 / 0.073 seconds
Next Scheduled Check: 09-19-2026 10:49:16
Last State Change: 09-19-2026 06:57:16
Last Notification: 09-19-2026 06:57:17 (notification 0)
Is This Service Flapping? NO (0.00% state change)
In Scheduled Downtime? NO
Last Update: 09-19-2026 10:48:25 (0d 0h 0m 3s ago)
Active Checks: ENABLED
Passive Checks: ENABLED
Obsessing: ENABLED
Notifications: ENABLED
Event Handler: ENABLED
Flap Detection: ENABLED

Service Commands

✗ Disable active checks of this service

🕒 Re-schedule the next check of this service

❓ Submit passive check result for this service

✗ Stop accepting passive checks for this service

✗ Stop obsessing over this service

✗ Disable notifications for this service

📧 Send custom service notification

🕒 Schedule downtime for this service

✗ Disable event handler for this service

✗ Disable flap detection for this service

✗ Clear flapping state for this service

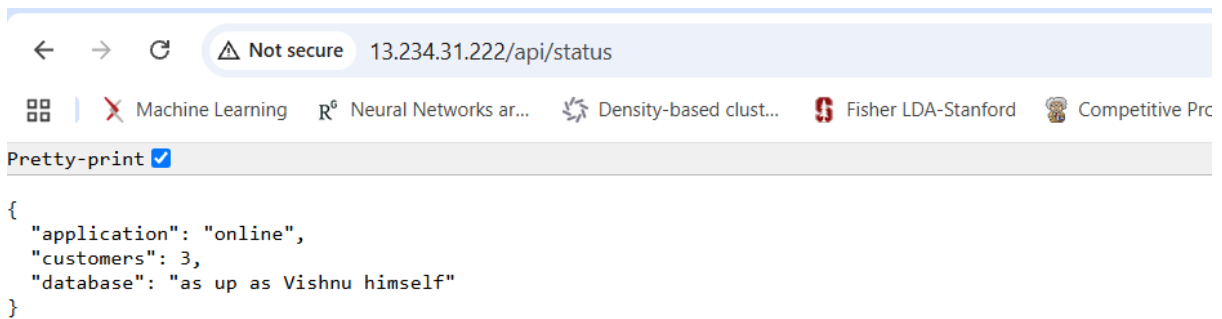
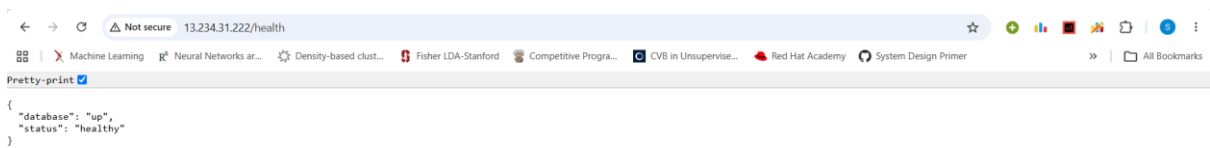
Service Comments

🗨 Add a new comment🗑 Delete all comments

Entry Time Author Comment Comment ID Persistent Type Expires Actions

This service has no comments associated with it

Page Tour



Stress Test:

```
ubuntu@nagios-mega: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/remote-linux.cfg *  
check_period                24x7  
notifications_enabled       1  
notification_period         24x7  
notification_options        w,c,u,r  
contact_groups              admins  
}  
  
define service {  
    use                      generic-service  
    host_name                 WEB-01  
    service_description       Web Application Health  
    check_command              check_web_health!http://172.31.8.87/health!0!0  
    check_interval            1  
    retry_interval            1  
    max_check_attempts        3  
    check_period              24x7  
    notifications_enabled     1  
    notification_period       24x7  
    notification_options      w,c,u,r  
    contact_groups            admins  
}
```

Service Information

Last Updated: Sat Sep 19 10:54:34 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin

Service
Web Application Health
On Host
Production Web Server
(WEB-01)
Member of
No servicegroups.
172.31.8.87

Service State Information

Current Status: **CRITICAL** (for 0d 0h 0m 9s)
Status Information: CRITICAL: Health check passed, response=0.02s
Performance Data:
Current Attempt: 1/3 (SOFT state)
Last Check Time: 09-19-2026 10:54:25
Check Type: ACTIVE
Check Latency / Duration: 0.001 / 0.155 seconds
Next Scheduled Check: 09-19-2026 10:55:26
Last State Change: 09-19-2026 10:54:25
Last Notification: N/A (notification 0)
Is This Service Flapping? **NO** (0.00% state change)
In Scheduled Downtime? **NO**
Last Update: 09-19-2026 10:54:33 (0d 0h 0m 1s ago)

Active Checks: **ENABLED**
Passive Checks: **ENABLED**
Obsessing: **ENABLED**
Notifications: **ENABLED**
Event Handler: **ENABLED**
Flap Detection: **ENABLED**

Service Commands

- ✗ Disable active checks of this service
- ⌚ Re-schedule the next check of this service
- ❓ Submit passive check result for this service
- ✗ Stop accepting passive checks for this service
- ✗ Stop obsessing over this service
- ✗ Disable notifications for this service
- ⌚ Delay next service notification
- 📧 Send custom service notification
- 🕒 Schedule downtime for this service
- ✗ Disable event handler for this service
- ✗ Disable flap detection for this service
- ✗ Clear flapping state for this service

Service Comments

⚡ Add a new comment ⌂ Delete all comments

Entry Time	Author	Comment	Comment ID	Persistent	Type	Expires	Actions
This service has no comments associated with it							

Note: Health check passed but the response time threshold was above the modified threshold

Nagios

Service Information
 Last Updated: Sat Sep 19 10:57:04 UTC 2026
 Updated every 90 seconds
 Nagios® Core™ 4.5.10 - www.nagios.org
 Logged in as nagiosadmin

Service
Web Application Health
 On Host
Production Web Server
(WEB-01)

Member of
No servicegroups.

172.31.8.87

Service State Information

Current Status: **CRITICAL** (for 0d 0h 0m 38s)
Status Information: CRITICAL: Health check passed, response=0.02s
Performance Data: 3/3 (HARD state)
Current Attempt: 09-19-2026 10:56:26
Last Check Time: 09-19-2026 10:56:26
Check Type: ACTIVE
Check Latency / Duration: 0.000 / 0.150 seconds
Next Scheduled Check: 09-19-2026 10:57:26
Last State Change: 09-19-2026 10:56:26
Last Notification: 09-19-2026 10:56:26 (notification 1)
Is This Service Flapping? **NO** (6.25% state change)
In Scheduled Downtime? **NO**
Last Update: 09-19-2026 10:57:02 (0d 0h 0m 2s ago)

Service Commands

- ✗ Disable active checks of this service
- ⌚ Re-schedule the next check of this service
- 🔍 Submit passive check result for this service
- ✗ Stop accepting passive checks for this service
- ✗ Stop obsessing over this service
- ✓ Acknowledge this service problem
- ✗ Disable notifications for this service
- ⌚ Delay next service notification
- 📧 Send custom service notification
- 🕒 Schedule downtime for this service
- ✗ Disable event handler for this service
- ✗ Disable flap detection for this service
- ✗ Clear flapping state for this service

Service Comments
 Add a new comment Delete all comments
 Entry Time Author Comment Comment ID Persistent Type Expires Actions
 This service has no comments associated with it

**Note: Notification sent upon reaching max_check_attempts
 3(let's check it)**

Gmail

in:spam

1 of 142

**** PROBLEM Service Alert: Production Web Server/Web Application Health is CRITICAL **** Spam

from: sreevishnu0407@gmail.com via amazonses.com
 to: me

16:26 (1 minute ago)

Why is this message in spam? This message is similar to messages that were identified as spam in the past.
 Report as not spam

***** Nagios *****

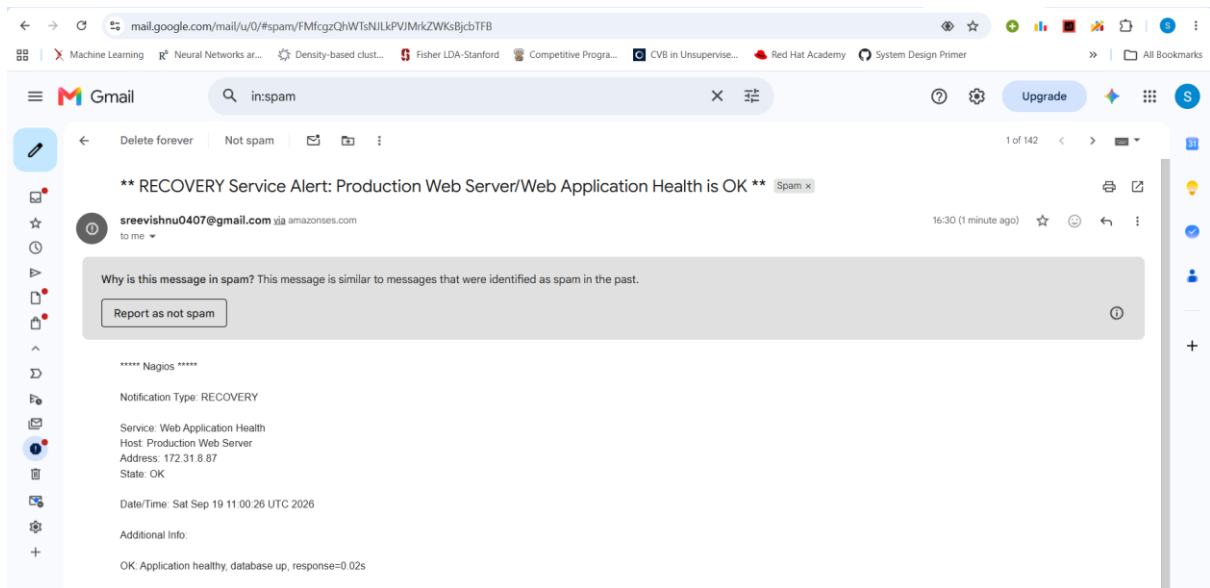
Notification Type: PROBLEM

Service: Web Application Health
 Host: Production Web Server
 Address: 172.31.8.87
 State: CRITICAL

Date/Time: Sat Sep 19 10:56:26 UTC 2026

Additional Info:
 CRITICAL: Health check passed, response=0.02s

Note: There we go, gotcha!!



Note: There we go successful completion of this extensive practical